

Quantifying heterogeneous stress distribution at the thin-section scale: Application of EBSD-based subgrain-size piezometry



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Research Motivation:

Continental shear zones are mechanically heterogeneous, but the magnitude and spatial scale of stress partitioning remain poorly constrained. Using the Vaikrita Thrust zone, we quantify stress and infer rheological behaviour at the thin-section scale to determine how lithological variations and mixed-phase micro-shear zones generate and localize rheological heterogeneity in a major continental shear system.

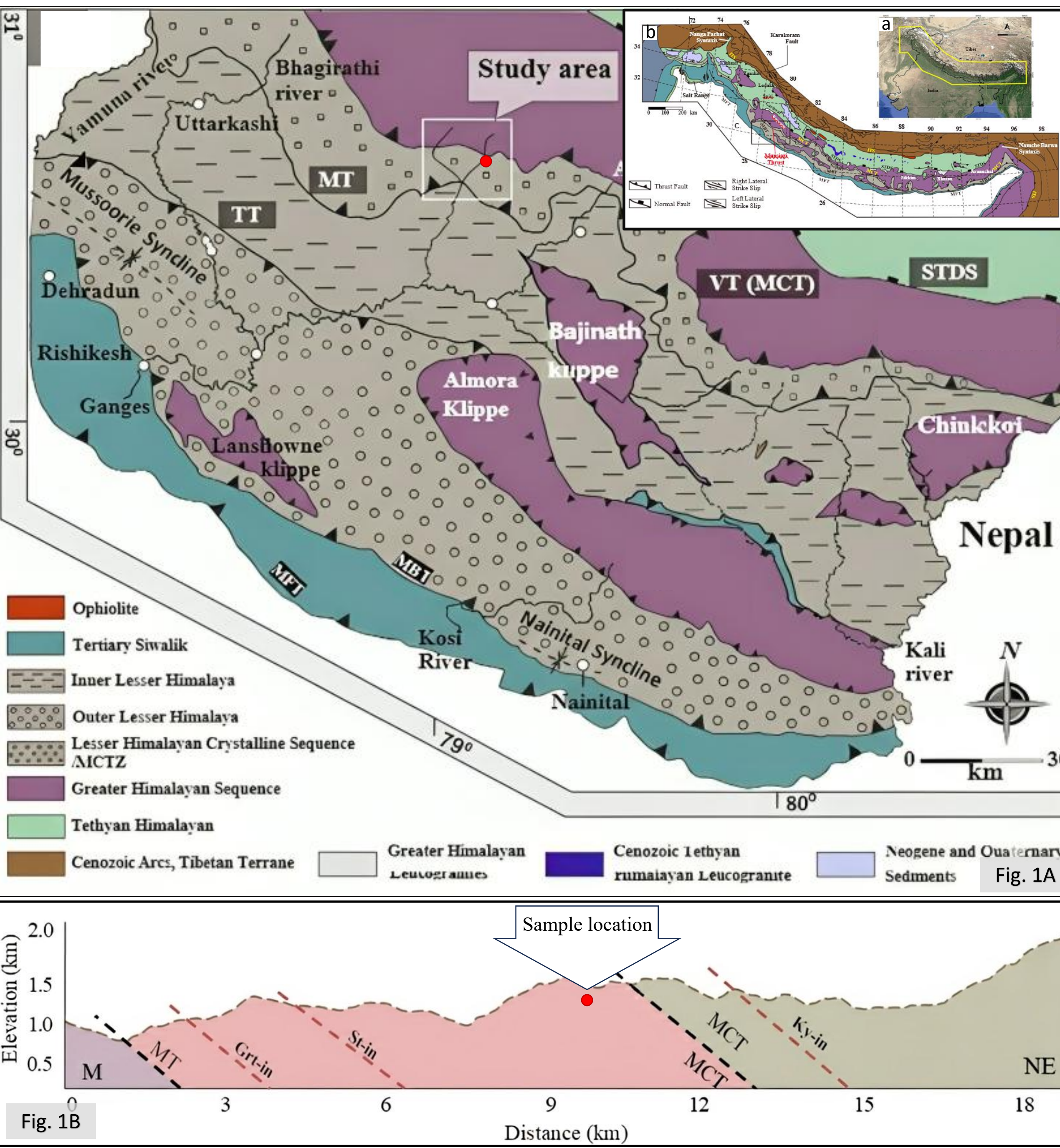
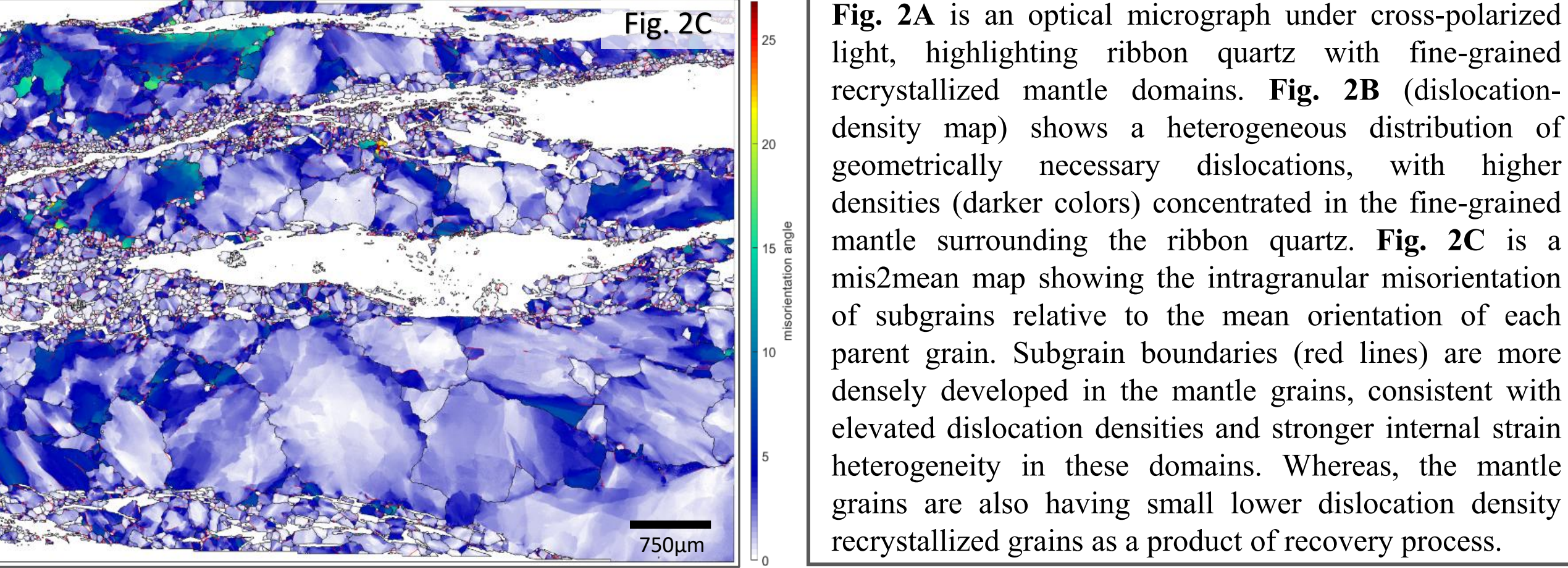
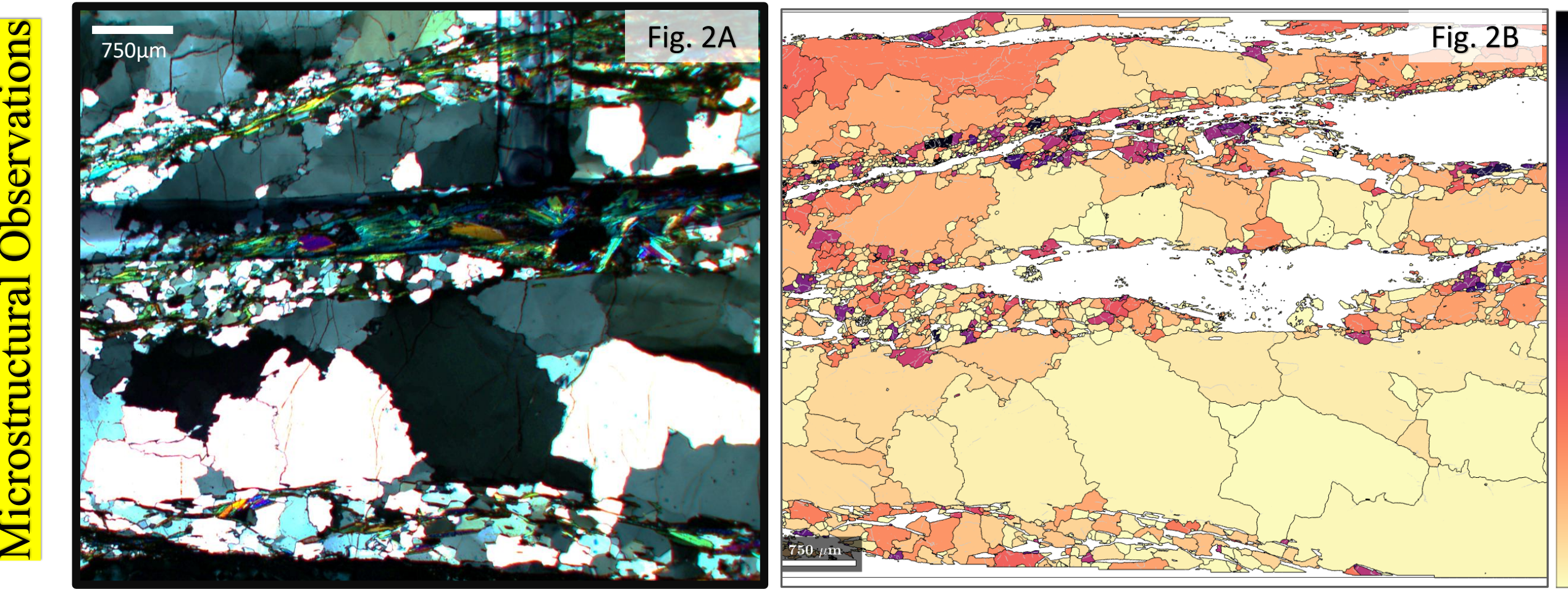


Fig. 1A. Geological and structural framework of the Kumaon-Garhwal Himalaya. (a) Location of the study area within the Himalayan orogen (yellow box). (b) Regional geologic map of the Central Himalaya (modified after Yin, 2006) showing major lithotectonic units and the study area. The sample was collected from the Vaikrita Thrust (VT/MCT) zone, between the Munsiri Thrust (MT) and the South Tibetan Detachment System (STDS). The red circle marks the sample site within the Outer Lesser Himalayan crystalline sequence. **Fig. 1B.** Simplified NE-SW geological cross-section (modified after Kohn, 2014). The sample location is projected onto the section.



Methodology:

1. Microstructural & Petrographic Analysis

Optical Microscopy is utilized to identify different minerals present and their shape, size optical properties under polarizing light. Specifically undulous extinction, chessboard extinction (Passchier & Trouw, 2005) (Fig. 2A, Fig. 2C) of quartz grains and lobate grain boundaries says about their deformation conditions by Grain Boundary Migration (GBM) (Stipp et al., 2002) (Fig. 2A).

2. EBSD Mapping & Fabric Analysis

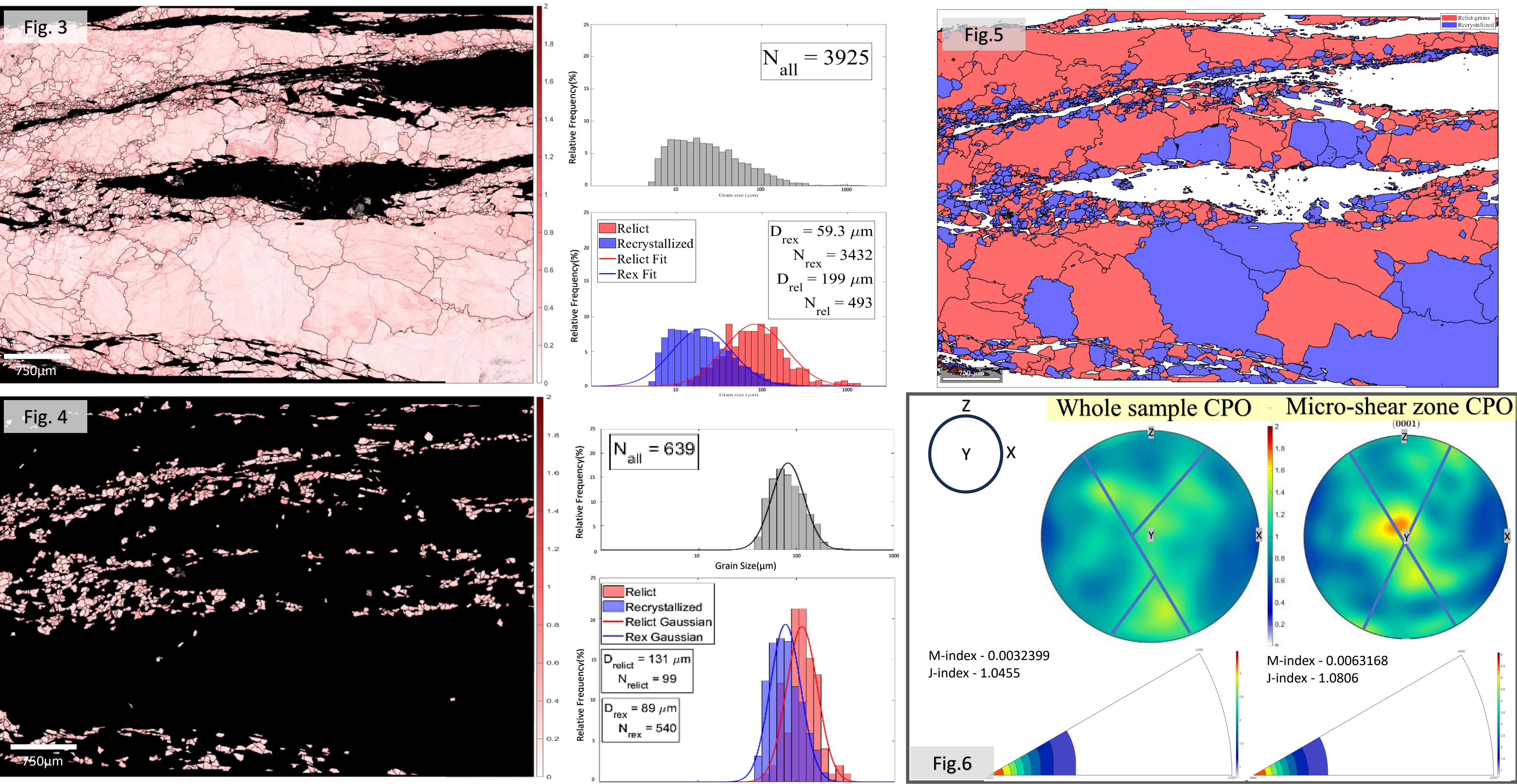
Electron Backscatter Diffraction (EBSD) is used to understand crystallographic deformation process with visual representation using MTEX 6.0.0 toolbox (<https://mtex-toolbox.github.io/>) for MATLAB (2024b) software (Bachmann et al., 2010). Kernal Average Misorientation (KAM) map (Fig. 3), recrystallized and relict grains map (Fig. 5), Orientation map (Fig. 7) are generated from EBSD data. Generated dislocation density (Fig. 2B) maps and mis2mean (Fig. 2C) maps to quantify intragranular misorientations and subgrain boundary networks. Crystallographic Preferred Orientations (CPO) are plotted to distinguish between bulk strain and localized shear zone strain.

3. Geothermometry

Peak metamorphic conditions are calculated using Garnet-Biotite thermometry (Holdaway, 2000): $633 \pm 25^\circ\text{C}$ and Garnet-Ilmenite thermometry (Martin et al., 2009): Yielding $655.3 \pm 57^\circ\text{C}$. Additionally high-temperature deformation ($>600^\circ\text{C}$) is corroborated by the microstructural observation of GBM and Subgrain Rotation (SGR) (Stipp et al., 2002).

4. Subgrain-Size Piezometry

Applied EBSD-based subgrain size piezometry to estimate differential stress from the mean subgrain size (λ) (Goddard et al., 2020). Low angle boundaries are defined within a range of 2° – 10° misorientation between neighboring pixels, where Grain boundaries are those at an angle $>10^\circ$. Stress is calculated separately for thick monophase quartz layers, thin ribbon quartz layers and bulk sample. RMS Grain Size Piezometry following Cross et al., 2017 is conducted for comparison between calculated stress values for bulk sample.



CPO patterns:

- Where the bulk sample displays a Type-I asymmetric crossed girdle characteristic of non-coaxial deformation, with basal $\langle a \rangle$ and rhomb $\langle a \rangle$ slip, while quartz within micro-shear zones (Fig. 4) show a weakly constructed Type-II crossed girdle (Passchier & Trouw, 2005) (Fig. 6).
- The alignment of c-axes sub-parallel to the maximum shear direction within the mixed domains (micro-shear zone) confirms that prism $\langle a \rangle$ slip is the primary operative slip system during localized shearing.
- In quartz, this concentration at the [c] poles in the IPF typically confirms that the deformation is being accommodated by specific lattice planes aligning with the shear direction which occurs at very high temperature conditions. The IPF for the Micro-shear zone shows a much tighter concentration around [c], confirming that the micro-shear zone experienced intense lattice rotation.

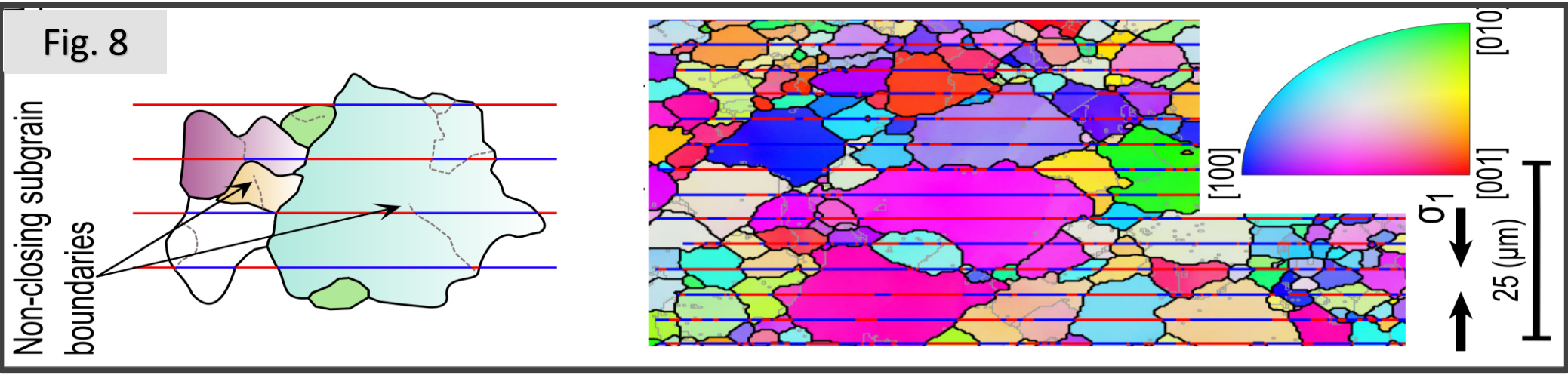
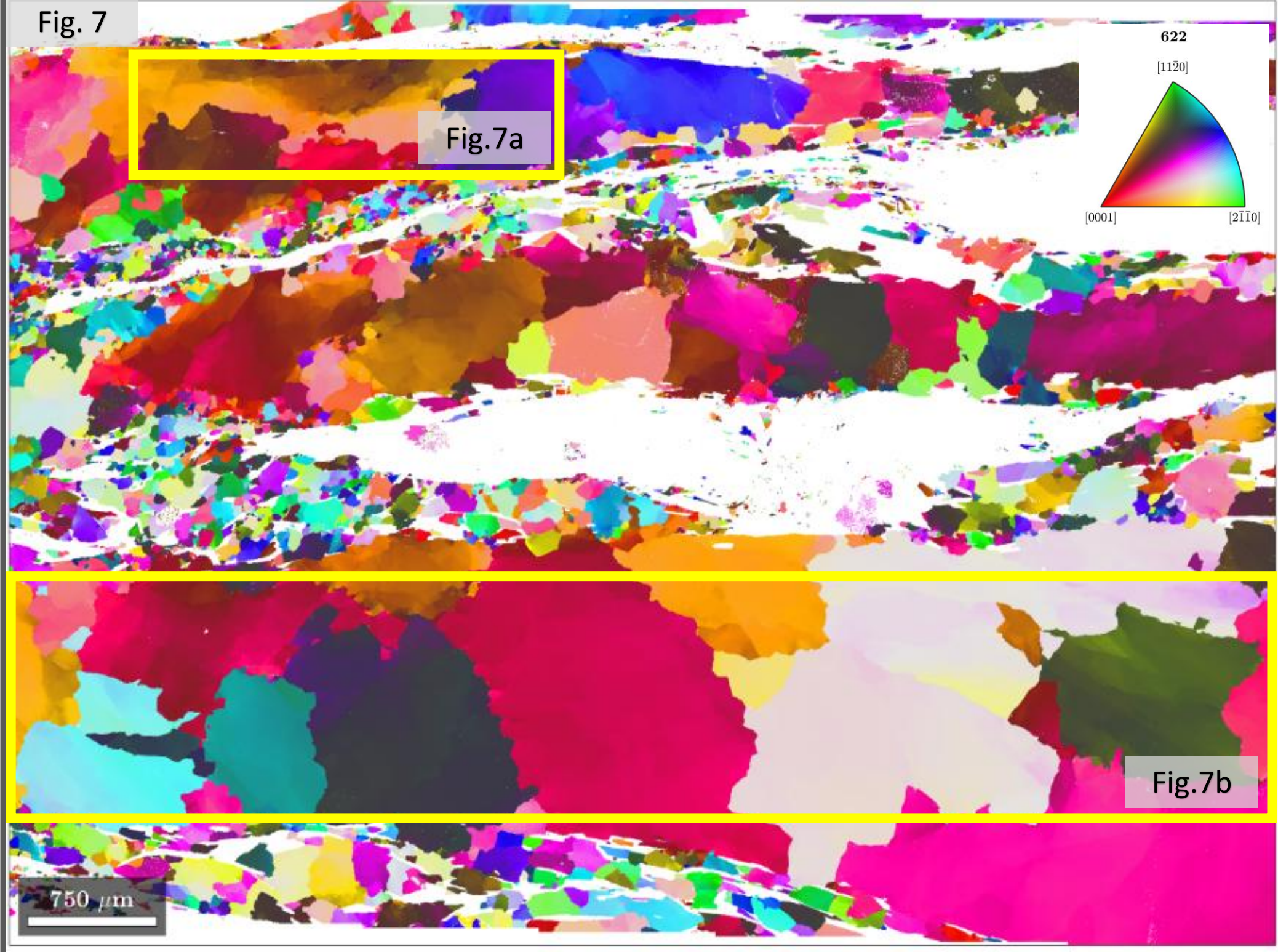
Discussion:

- GBM, SGR along with microstructural observations like chessboard extinction (Fig. 2C) confirms that the deformation took place at $>600^\circ\text{C}$. Thermometry yields peak temperatures of ~ 633 – 655°C , consistent with deformation at $>600^\circ\text{C}$.
- Linear intercept subgrain size piezometry method reveals stress partitioning in thin-section scale. Calculated stress is higher at the mixed domains (~ 8 MPa) (Fig. 7a) than the thick quartz domain (~ 3 MPa) (Fig. 7b). This thin layer with mixed domain is representing the bulk sample stress of ~ 9 MPa.
- Root-Mean-Square (RMS) recrystallized quartz grain size ($76 \mu\text{m}$) overestimates the paleostress if it is compared to linear intercept method, that is ~ 31 MPa for the bulk sample.
- Stress data is converted to shear stress following Paterson & Olgaard, 2000 and plotted against temperature and compared with Goddard, C. R. (2021) (Fig. 9). Also, depth is estimated of about 20 Km (Fig. 10).
- The transition to a weak Type-II crossed girdle within micro-shear zones suggests the influence of secondary phases (e.g., mica). These phases likely acted as pinning agents at grain boundaries (Smith, 1948), inhibiting grain boundary migration (GBM) and leading to Subgrain Rotation (SGR). It also shifts the deformation regime; rather than strain being accommodated solely by dislocation creep in quartz, it is distributed across multiple phases, evidenced by lower J-index.
- The presence of secondary minerals facilitates heterogeneous strain distribution and stress partitioning, governing the rheological evolution of the Vaikrita Thrust during high-temperature deformation.

Literature Study: 1. Goddard, R. M., Hansen, L. N., Wallis, D., Stipp, M., Holyoke III, C. W., Kumamoto, K. M., & Kohlstedt, D. L. (2020). A subgrain-size piezometer calibrated for EBSD. *Geophysical Research Letters*, 47(23), e2020GL090056. 2. Goddard, C. R. (2021). *Subgrain-size piezometry as a tool for measuring stress in polycrystalline rocks* (Doctoral dissertation, University of Oxford). 3. Cross, A. J., Prior, D. J., Stipp, M., & Kidder, S. (2017). The recrystallized grain size piezometer for quartz: An EBSD-based calibration. *Geophysical Research Letters*, 44(13), 6667–6674. 4. Shivaji, S. (2020). Tectono-metamorphic evolution of the Himalayan metamorphic rocks: insights from the Mandakini and Madhmaheswar Ganga river valley, Northwestern India (Doctoral dissertation, 北海道大学).

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Subgrain-size piezometry (via Goddard et al. 2020)



	Arithmetic mean of linear intercept (λ) (μm)	Calculated stress (Goddard et al. 2020) (MPa)
7a	55.25	8.46
7b	202	2.87
Whole sample	59.38	9.29

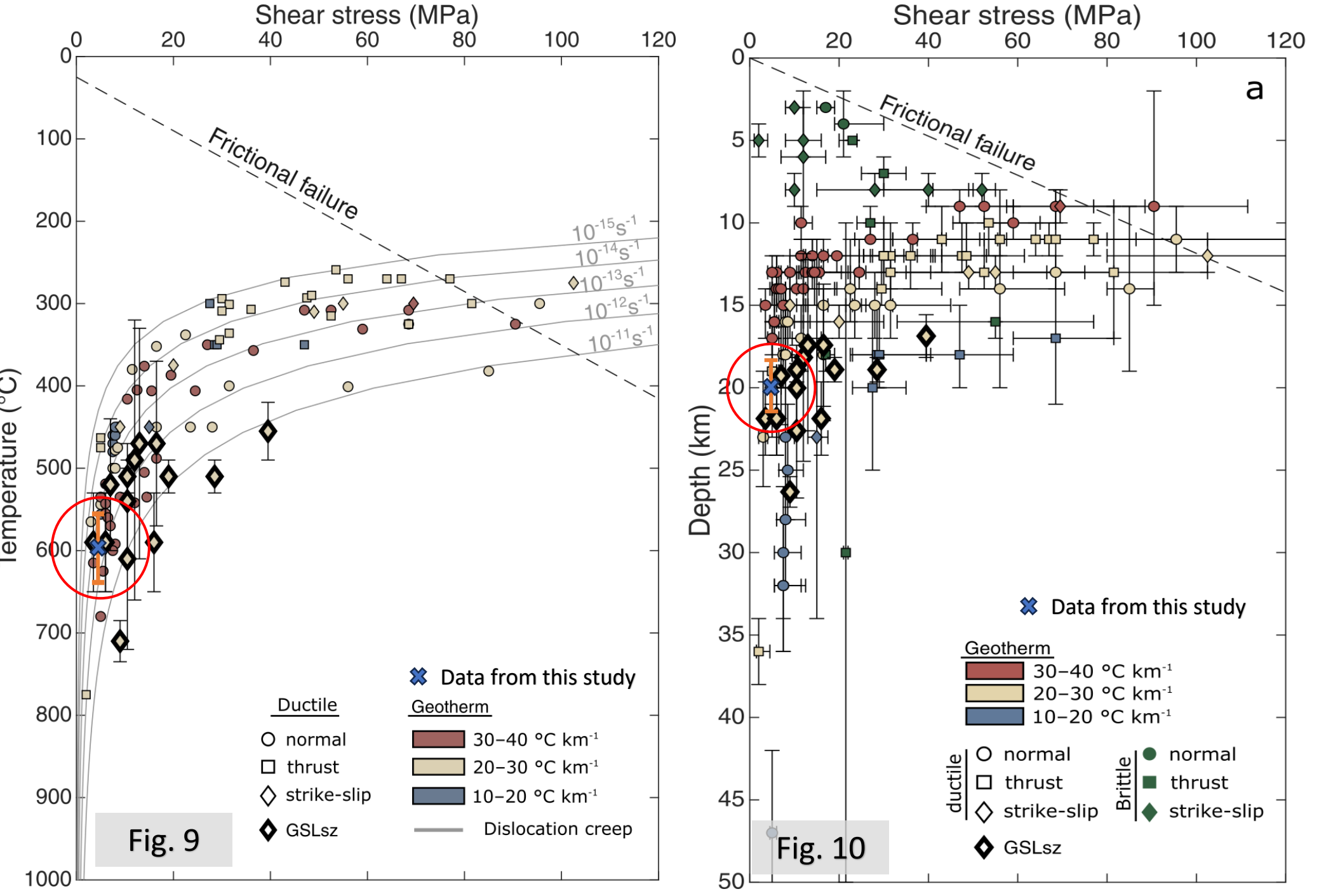


Fig. 8. The line-intercept method (Goddard et al., 2020). **Fig. 9.** Stress data from bulk sample is converted to shear stress and has been plotted on temperature and Depth versus Shear Stress plots. (Goddard, C. R., 2021).